

Interface Inheritance in Java

Basic Information

Assignment: Assignment 2

Program: InterfaceInheritanceDemo.java

Topic: Interfaces — Different Types of Implementations of Interfaces

Objective / Aim

To demonstrate inheritance between interfaces using the extends keyword and implementation of the child interface by a class.

System Requirements

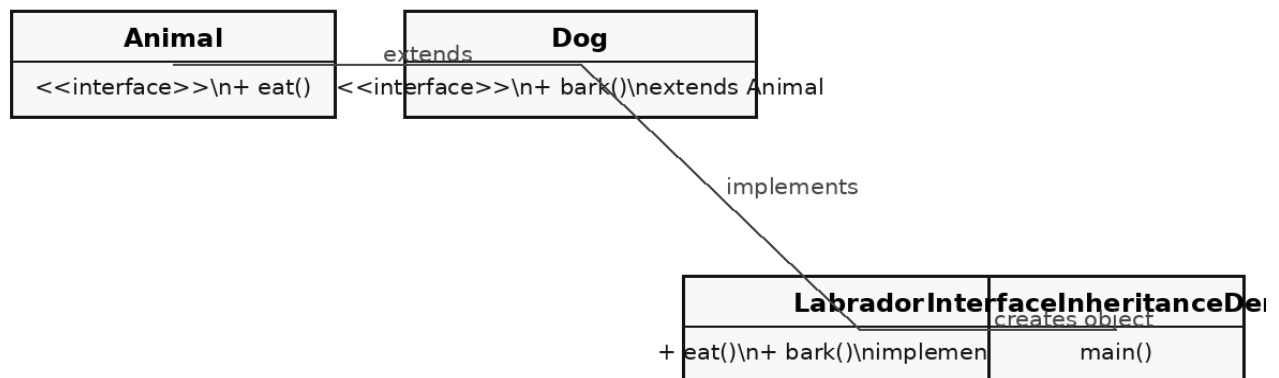
Hardware: Computer/Laptop, minimum 4 GB RAM, keyboard and mouse.

Software: Java Development Kit (JDK), Java compiler, Java Runtime Environment (JRE), and any Java-supported IDE or terminal.

Algorithm / Logic Description

1. Start the program.
2. Create the parent interface Animal with the eat() method.
3. Create the Dog interface that extends Animal and declares bark().
4. Create Labrador that implements Dog.
5. Define eat() and bark() in Labrador.
6. Create a Labrador object in main().
7. Call eat() and bark() using the object.
8. Stop the program.

UML Class Diagram



Source Code

```
// Parent interface
interface Animal {

void eat();
}

// Child interface inherits Animal
interface Dog extends Animal {

void bark();
}

// Class implements the child interface
class Labrador implements Dog {

public void eat() {
System.out.println("Labrador is eating");
}

public void bark() {
System.out.println("Labrador is barking");
}
}

public class InterfaceInheritanceDemo {
public static void main(String[] args) {

Labrador dog = new Labrador();

dog.eat();
dog.bark();
}
}
```

Compilation & Execution Commands

Save as: InterfaceInheritanceDemo.java

Compile: `javac InterfaceInheritanceDemo.java`

Execute: `java InterfaceInheritanceDemo`

Output Screen

```
C:\Java\Assignment2> java Program  
Labrador is eating  
Labrador is barking
```

Result

Thus, the interface inheritance program was implemented successfully using the interface concepts demonstrated in the source code.